

# Linear Programming - Duality and Geometry

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# Overview

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## Definition of a Linear Program

An optimization is a linear programming if both the objective function and constraints are linear in the argument  $x \in \mathbb{R}^n$

## Structure of a general linear program

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^T x \\ \text{subject to} \quad & Ax \leq b \\ & A \in \mathbb{R}^{m \times n} \end{aligned}$$

**NOTE:** An equality constraint, say  $z = 3$  can be written as two inequality constraints  $z \leq 3$ ,  $-z \leq 3$

# Linear Program in Standard Form

A linear program in standard form refers to the following problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^T x \\ \text{subject to} \quad & Ax = b, x \geq 0 \\ & A \in \mathbb{R}^{m \times n} \end{aligned}$$

Any linear program can be converted into standard form:

- $Ax \leq b$  is the same as  $Ax - b = z$  and  $z \geq 0$  (inequality  $\rightarrow$  equality and a non-negativity by additional slack variable  $z$ )
- $x = x^+ - x^-$  where  $x^+, x^- \geq 0$  and

$$c^T x = [c^T \quad -c^T \quad 0] \begin{bmatrix} x^+ \\ x^- \\ z \end{bmatrix}$$

$$\begin{bmatrix} A & -A & I \end{bmatrix} \begin{bmatrix} x^+ \\ x^- \\ z \end{bmatrix} = b \implies Ax + z = b, z \geq 0$$

# Optimality Conditions

Writing the Lagrange function - multiplier:  $\lambda, s$

$$L(x, \lambda, s) = c^T x - \lambda^T (Ax - b) - s^T x$$

**Optimality (KKT) Conditions** - Necessary and Sufficient:

*Gradient Condition:*  $A^T \lambda + s = c$

*Feasibility:*  $Ax = b$

*Feasibility:*  $x \geq 0$

*Inequality:*  $s \geq 0$

*Complementary Slackness:*  $x_i s_i = 0$

# Rewriting the KKT conditions - Towards Duality

- Let  $(x^*, \lambda^*, s^*)$  be the optimal point. Then by KKT conditions,

$$c^T x^* = (A^T \lambda^* + s^*)^T x^* = \lambda^{*T} (Ax^*) = \lambda^{*T} b$$

- Let  $\bar{x}$  be any other feasible point - i.e.  $\bar{x} \geq 0, A\bar{x} = b$ . Then,

$$c^T \bar{x} = (A^T \lambda^* + s^*)^T \bar{x} = b^T \lambda^* + \bar{x}^T s^* \geq b^T \lambda^* = c^T x^*$$

- So, we have that  $\bar{x}$  is optimal if and only if

$$\bar{x}^T s^* = 0$$

# Dual Problem

- **Primal Problem:**

$$\begin{aligned} & \min_{x \in \mathbb{R}^n} c^T x \\ & \text{subject to } Ax \leq b, x \geq 0 \end{aligned}$$

- **Dual Problem:**

$$\begin{aligned} & \max_{\lambda \in \mathbb{R}^m} b^T \lambda \\ & \text{subject to } A^T \lambda \leq c \end{aligned}$$

- **Primal  $\leftrightarrow$  Dual** (trick to remember):

$$x \leftrightarrow \lambda$$

$$c \leftrightarrow b$$

$$A \leftrightarrow A^T$$

# Optimality (KKT) conditions for the Dual Problem



$$\max_{\lambda \in \mathbb{R}^m} b^T \lambda = \min_{\lambda \in \mathbb{R}^m} -b^T \lambda$$

subject to  $s := c - A^T \lambda \geq 0$

- Lagrangian  $\bar{L}$  for the dual problem :  $x$ -Lagrange multiplier for the constraint  $s = c - A^T \lambda \geq 0$

$$\bar{L}(\lambda, x) = -b^T \lambda - x^T (c - A^T \lambda)$$

- **Optimality (KKT) conditions for the Dual problem:**

*Gradient Condition:*  $Ax = b$

*Feasibility:*  $s := c - A^T \lambda \geq 0$

*Non-negativity:*  $x \geq 0$

*Complementary Slackness:*  $x_i s_i := x_i (c - A^T \lambda)_i = 0$

*Definition of  $s$ :*  $s = c - A^T \lambda$

# Identical conditions

- Observe the primal and dual KKT conditions side by side

*Gradient Condition:*  $A^T \lambda + s = c$

*Feasibility:*  $Ax = b$

*Feasibility:*  $x \geq 0$

*Inequality:*  $s \geq 0$

*Complementary Slackness:*  $x_i s_i = 0$

**Primal KKT**

*Gradient Condition:*  $Ax = b$

*Feasibility:*  $s := c - A^T \lambda \geq 0$

*Non-negativity:*  $x \geq 0$

*Complementary Slackness:*  $x_i s_i := x_i (c - A^T \lambda)_i = 0$

*Definition of  $s$ :*  $s = c - A^T \lambda$

**Dual KKT**

- They are identical - except that the titles are different!

# Strong Duality Theorem

## Strong Duality Theorem

- If the primal/dual problem is solvable , then so is the other dual/primal problem
- If the primal/dual problem is unbounded , then the other dual/primal problem is infeasible

### Proof:

- Let  $(x^*, \lambda^*, s^*)$  have a solution for the primal/dual problem. But the KKT conditions are identical for both.
- Let the primal problem be bounded - ie.  $\exists x_k$  such that  $c^T x_k \rightarrow -\infty$  satisfying  $Ax_k = b$  and  $x_k \geq 0$ .
- Let the dual be feasible with  $\bar{\lambda}$  such that  $A^T \bar{\lambda} \leq c$ . Then,

$$\bar{\lambda}^T b = \bar{\lambda}^T (Ax_k) \leq c^T x_k$$

which is a contradiction as  $k \rightarrow \infty$

# Geometry of the Problem

Consider a problem  $\min c^T x$  in  $\mathbb{R}^2$  with five inequality constraints  $Ax \leq b$ , i.e.  $A \in \mathbb{R}^{5 \times 2}$

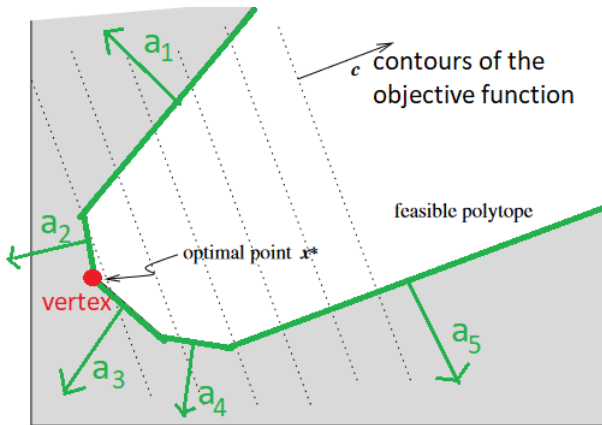
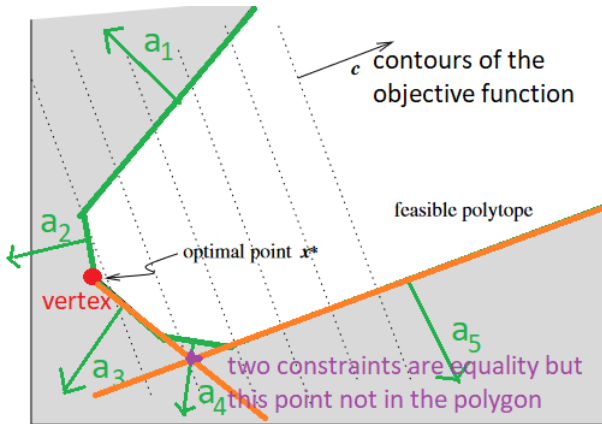


Figure: Optimal point is obtained typically at a vertex

- One sees that the feasible set defined by the constraint set  $Ax \leq b$  is a convex polygon in  $\mathbb{R}^2$
- We see that typically as we move along the black dotted contours of  $c^T x$  to minimize it, the minimum is hit at a vertex typically - if  $c$  is parallel to one of the  $a_i$ , then the minimum is obtained at a full edge but still the vertices being a part of the edge, is a minimizer
- What is mathematically a vertex? It is intersection of any two green lines - where the inequality becomes equality
- More precisely, a vertex of a polygon in  $\mathbb{R}^2$  is a **feasible point** where any **two of the inequality constraints are satisfied with equality**

# Importance of feasibility condition

In the definition of a vertex, feasibility is important as well. As an example,



**Figure:** A vertex should have two equality constraint and feasible as well - the violet point is not a vertex as it is not feasible (not inside the polygon)

# Vertex: From $\mathbb{R}^2$ to a general LP

- Consider the standard form LP<sup>1</sup>

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^T x \\ \text{subject to} \quad & Ax = b, x \geq 0 \\ & A \in \mathbb{R}^{m \times n}, \text{ full rank with } m < n \end{aligned}$$

- The set of solutions to  $Ax = b$  will form a  $n - m$  dimensional space since rank of  $A$  is  $m$ . ( $n$  variables and  $m$  equations  $\Rightarrow n - m$  free variables)
- In this  $n - m$  dimensional feasible set, one needs  $n - m$  equations to give a unique point.
- In  $\mathbb{R}^2$ , a vertex was defined by two inequality constraints being satisfied with equality. In this  $n - m$  dimensional space, a vertex should hence be defined by  $n - m$  inequality constraints being satisfied with equality

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<sup>1</sup>If  $m > n$ , then there exists only one or no feasible point in which case the problem is trivial

# Basic Feasible Point: Extension of a vertex

The notion of a **basic feasible point** is the extension of the definition of a vertex

## Basic Feasible Point

A point  $x$  is called a **basic feasible point** w.r.t a standard form LP if

- $x$  is feasible -  $Ax = b, x \geq 0$
- there exists  $m$  indices  $\mathcal{B} := \{i_1, i_2, \dots, i_m\} \subset \{1, 2, \dots, n\}$  such that
- 

$$x_i = 0 \text{ if } i \notin \mathcal{B}$$

- The  $m \times m$  matrix  $B$  defined by  $B := [A_i]_{i \in \mathcal{B}}$  where  $A_i$  is the  $i^{\text{th}}$  column of  $A$  is non-singular<sup>a</sup>

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<sup>a</sup>As a consequence, the set of  $n - m$  equations  $x_i \geq 0$  for  $i \notin \mathcal{B}$  and the  $m$  equations  $Ax = b$  has a unique solution which is the basic feasible point  $x$

# Theorem of solvability by basic feasible points

It turns out that under just trivial conditions, LP problems can be solved by just basic feasible points

## Fundamental Theorem of LP

- If the LP is feasible and bounded, then it has atleast one optimal solution  $x^*$
- Atleast one point in the optimal solution is a basic feasible point
- If the LP is feasible, then it has basic feasible points

- As a consequence, it is enough to solve

$$\begin{aligned} \min_{x \in \mathfrak{B}} c^T x \\ x \geq 0, \quad Ax = b \end{aligned}$$

where  $\mathfrak{B}$  is the set of all basic feasible points

- But the set of basic feasible points is finite as subsets of  $\{1, 2, \dots, n\}$  are finite and each such subset  $\mathcal{B}$  generates a unique basic feasible point - so it is now a search among finitely many points!